

PROBLEM STATEMENT REVIEW

Product Search and Recommendation System for Shopping

TEAM 4

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Abstract

The Product Search and Recommendation System for Shopping is designed to provide a smarter and more efficient way for users to search and discover products on an online shopping platform. With the rapid growth of e-commerce, customers often find it difficult to locate the right product due to incorrect spellings, incomplete product names, or a large number of similar products. The proposed system overcomes these challenges by enabling users to search using product names, categories, brands, keywords, or descriptive queries. It retrieves the most relevant products from a structured product repository and presents them in an organized and user-friendly manner, making the search process faster and more accurate.

In addition to efficient product search, the system recommends similar or related products based on their category, brand, features, and user preferences. It also allows users to apply filters such as price range, brand, ratings, and availability to narrow down their search results and make informed purchasing decisions. By combining intelligent search with personalized recommendations, the system enhances the overall shopping experience, reduces the time spent searching for products, and helps users easily discover products that best match their needs. The proposed solution is scalable, reliable, and suitable for modern e-commerce applications that manage large product collections.

02

Problem Statement

Modern e-commerce platforms host millions of products, making it hard for customers to quickly find what they need. Traditional search fails on incomplete names, typos, or natural-language queries—leading to irrelevant results, wasted time, and missed alternatives.

A real user search:

"laptop for programming under ₹60,000"

Massive Catalogs

Millions of listings make manual browsing impractical

Typos & Incomplete Names

Small errors return zero relevant results

Natural-Language Intent

Descriptive queries aren't understood by keyword search

Missed Alternatives

Users don't discover suitable similar products

03 Proposed Functionalities

Registration & Login

Secure user sign-up and authentication

Admin Product Control

Add, update, and delete product listings

Keyword & Query Search

Search by keywords or descriptive queries

Typo Tolerance

Handles spelling mistakes & incomplete names

Smart Filtering

By category, brand, price range, and ratings

Relevance Ranking

Results displayed by relevance score

Product Recommendations

Suggests similar or related products

Explainable Suggestions

Brief reasons given for each recommendation

Detailed Product View

Full specifications and information per item

Search History

Maintains a record of user searches

04 Tools and Technologies

PROGRAMMING LANGUAGE

Java

BACKEND

Java

IDE

IntelliJ IDEA / Eclipse / NetBeans

VERSION CONTROL

Git (optional)

FRONTEND

Java Swing / JavaFX

DATA STORAGE

Java Collections (ArrayList / HashMap)

DATABASE TOOL

File Handling

OPERATING SYSTEM

Windows / Linux

05 Expected Outcome

The system will deliver an intelligent, efficient platform for searching and recommending products from large repositories—finding results even with typos, incomplete names, or descriptive queries—while improving decision-making through meaningful, relevant suggestions.

Overall, the project reduces search time, improves accuracy, enhances user satisfaction, and demonstrates practical, advanced search techniques for modern e-commerce.

**Faster
Search**

**Higher
Accuracy**

**Smart
Recommendations**

**Better
Decisions**